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# Anchor, Don't Name: What an Ontology Actually Contributes to LLM-Built Knowledge Graphs

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## Abstract

Enterprises increasingly build knowledge graphs from their documents with LLM extractors, and in regulated domains such as finance the answers drawn from those graphs must be verifiable, not merely plausible. The standard prescription is to guide extraction with a domain ontology such as FIBO. Yet whether the ontology improves the graph — and whether the graph then improves answering — has rarely been measured with the schema as the only variable and memorization controlled. We measure both, in a pre-registered study on the FinDER financial-QA corpus: the same filings are extracted under five schema conditions, from no schema to full FIBO, by three LLMs, with the two decisive conditions replicated on 420 stratified cases. Three results follow. First, the ontology does not improve extraction: it lowers cross-model agreement on entity names and does not raise coverage of answer-bearing facts; its measurable benefit is that schema violations become detectable. Second, comparing facts by entity name — standard alignment practice — hides most cross-model disagreements: aligning each extracted value by its exact position in the source document instead finds  $1.7\text{--}2.8\times$  as many comparable facts and exposes that a quarter of anchored values carry unit-scale errors. Third, standard QA scoring favors source text over the serialized graph, but cannot distinguish answers grounded in the supplied evidence from answers recalled from pretraining: separating memorized answers, honest refusals, and grounded answers reverses the comparison on two of three models, and adding provenance pointers alone raises accuracy. We conclude that facts should be aligned by provenance rather than by name, that ontologies are better deployed as validators than as extraction guides, and that evaluations of retrieval structure must condition each verdict on what the supplied evidence could support. Three pre-registered hypotheses failed and are reported as findings.

## 1 Introduction

Enterprises that build knowledge graphs from documents with LLMs face a design decision early: whether to guide extraction with a domain ontology. The conventional answer is yes — a schema should make extraction more consistent, more complete, and easier to query. Financial services is the domain where this belief is most institutionalized: FIBO [1], an OWL ontology of financial concepts maintained by the EDM Council, exists precisely to standardize how institutions name and relate the entities in their filings.

This paper measures that belief end to end on FinDER [2], a financial question-answering dataset built from SEC filings, and reports three findings that revise it. One number previews all three. We ran three LLM extractors over the same filings and compared the numeric facts they produced whenever two extractors drew a number from the same position in the source document. Their extracted values disagreed 1,482 times — and **in all 1,482 cases the disagreement is the same kind**: the digits match, and the extractors differ only on whether a table header such as “in thousands”

41 applies. Matching facts by entity name hides every one of these conflicts; matching them by source  
42 position exposes them; and no ontology in the prompt prevents them.

43 Each finding is stated with its paired confidence interval in Section 3. First, the ontology does not  
44 improve extraction: it lowers cross-model agreement on entity names, covers fewer of the numbers  
45 the gold answers require, and what survives is a validation surface. Second, the field's standard  
46 alignment key — the entity name — hides the disagreements that matter, and every disagreement  
47 it hides turns out to be a unit-scale conflict that provenance exposes. Third, once memorization  
48 and honest abstention are separated from grounded correctness, whether the graph helps answering  
49 becomes a measurable property of the question stratum and the model, not of the representation  
50 alone.

51 The three findings compose into one conclusion. The first removes the assumed mechanism: the  
52 ontology does not standardize what extractors say. The second identifies the failure that actually  
53 needs catching — scale conflicts — and shows that the name-based alignment everyone builds on  
54 cannot see it, while source position can. The third shows the same blindness repeated at evaluation  
55 time: scoring that cannot see evidence cannot price structure at all. What remains standing in all  
56 three is the same primitive. Our central claim is therefore a single sentence: *the value of structure*  
57 *is not a better answer but a verifiable one, and verification needs provenance, not names.* Every  
58 registered claim above carried its disconfirming outcome before the run that tested it (Section 2);  
59 three registered hypotheses were disconfirmed and are reported as such, and exploratory findings  
60 are labelled and replicated under fresh registration.

## 61 2 Study design

62 Three samples cover FinDER's own annotation grid (balanced 280; arithmetic 140; a 204-case cen-  
63 sus of the reasoning stratum), extracted by three LLMs under five schema conditions, each chosen  
64 to isolate one belief about what guidance contributes: a schema-free floor (what agreement costs  
65 nothing), a hand-written 20-class schema (a small bespoke schema suffices), corpus-scoped FIBO  
66 — the 70 classes ranked by how often the corpus itself mentions them (the standard ontology, as  
67 used) — and FIBO plus synonyms and FIBO plus its class hierarchy (richer vocabulary, respectively  
68 structure, improves extraction). The conditions nest, so each contrast attributes to a single addition  
69 (Table 3), and everything except the schema is byte-fixed and hashed. Two instruments do the work  
70 the field usually delegates to names and gold answers. **Provenance-anchored alignment:** every ex-  
71 tracted value is located at the unique numeric token it came from in the source, so facts align across  
72 extractors by source position, not by what each model chose to call them. **Evidence-conditional**  
73 **evaluation:** every answer verdict is conditioned on whether the served evidence contains the gold  
74 numbers, mechanically separating grounded correctness from memorization and honest abstention  
75 from failure. Both are deterministic; an LLM-judge panel is secondary only, cross-judging (no  
76 model grades its own answers) and admitted only after passing must-fail calibration controls. Every  
77 uncertainty interval in this paper is a 95% confidence interval obtained by nonparametric bootstrap  
78 over *cases* (5,000 resamples, fixed seed), and every comparison between conditions is computed as  
79 a paired within-case difference before resampling, so per-case difficulty cancels; we call a difference  
80 *separated* when its interval excludes zero. Hypotheses were registered with their disconfirming out-  
81 comes before each run, and the registrations are committed files in the supplementary material. Full  
82 design, prompts, and protocol: Appendix B.

## 83 3 Findings

### 84 3.1 The ontology provides detectability, not better extraction

85 With everything but the schema fixed, ontology guidance *lowers* cross-model agreement on en-  
86 tity names (schema-free minus FIBO: +0.069, 95% CI [+0.046, +0.091], paired over 280 cases)  
87 and covers *fewer* answer-bearing numbers (−0.036 [−0.067, −0.008]) — the registered content-  
88 advantage hypothesis reversed at scale. What survives is a validation surface: 529–1,038 schema  
89 violations per condition become mechanically checkable where the schema-free graph offers almost  
90 nothing to check. The synonym condition never mattered because it could not: only 0.5% of the  
91 corpus's questions need vocabulary mapping, and the abbreviations they use are analyst shorthand,  
92 not FIBO's. The full five-condition dose–response, with intervals, is Table 4 in Appendix C.

### 3.2 Extractors never disagree about digits — only about scale

Aligning facts by anchor instead of name yields  $1.7\text{--}2.8\times$  as many comparable cross-model facts, and 80–84% of the disagreements it exposes are invisible to any name key (registered replications on both conditions at 280 cases). Auditing those disagreements produced this study's sharpest fact: all 1,482 of them are unit-scale splits. A representative case, verbatim from the artifacts: at one source token, one extractor stored the fact "*revenue software licenses*" = 945,797 while another stored "*software licenses revenue fy2021*" = 945,797,000 — same digits, a  $10^3$  split from a table's "in thousands" header, and two names no string matcher would pair, so the conflict is invisible to name-keyed pipelines and would surface only as an unexplained thousand-fold error in whichever answer it reaches. Overall, 26.7% of anchored values match their source only after rescaling by  $10^3$  or  $10^6$ . Two deployment consequences follow: name-keyed entity resolution silently discards the verification signal a second extractor was supposed to buy, and cross-view verification has a hard ceiling regardless of key — 87% of facts are held by one view only, a share that *rises* with scale, and agreement-gating trades four right answers per wrong answer blocked (registered, disconfirmed; Appendix D).

### 3.3 Whether the graph helps answering depends on the question stratum and the model

Standard scoring says gold passages beat the serialized graph. But closed-book performance spans 0.06–0.31 across models on the same public filings, so a correct answer is weak evidence the evidence was used. Decomposing every verdict by whether the served evidence could support it (Table 2) reverses the reading on the prose stratum — the graphs' grounded-correct yield matches or beats passages on two of three models, passages carry  $2\text{--}4\times$  the memorization, and the third model's deficit is over-refusal, not misunderstanding. On the arithmetic stratum the reversal does *not* replicate: computing a value needs co-located operands that extraction fragments, and the original graph-favorable hypothesis closed negative on its own chosen ground. Attaching provenance pointers — position references only, no text — raised accuracy on two of three models and repaired the over-refusing model's deficit (Appendix E).

The third stratum completes the argument and adds the study's only graph-over-text separation. On the 204-case census of questions FinDER annotates as requiring reasoning — a stratum the models barely memorize (closed book 0.07–0.15) and where a pilot measurement showed the highest share of cross-extractor-verifiable facts (0.80 vs. 0.44 on the balanced sample) — at least one serialized-graph condition beats gold passages with separated intervals on every model (FIBO graph  $-0.060$  [ $-0.117, -0.004$ ] on gpt-oss; schema-free graph  $-0.055$  [ $-0.110, -0.002$ ] on MiniMax; and on DeepSeek — the model that loses with graphs on every other stratum — *both* graph conditions separate above passages,  $-0.085$  and  $-0.131$ , with provenance pointers its best condition. A prospective prediction registered before this experiment ran — that each model's grounded-correct lift would exceed its balanced-sample lift — landed two of three and is reported as failed under its own all-models clause: gpt-oss  $+4 \rightarrow +10$  and DeepSeek  $-15 \rightarrow +2$  hit, MiniMax  $+14 \rightarrow +6$  missed, compressed against a ceiling (passages already grounding 148 of 169 cases). The profile's directional signal coincided with the reversal on all three models; its registered predictive form did not survive its first prospective test.

## 4 A pre-deployment verifiability profile

Each design decision this study evaluates pays off only under measurable preconditions, all computable from one pilot extraction: the verifiable share (13% here), the name-blind disagreement rate (80–84%), the scale-fragility rate (26.7%), the guardrail surface, per-model legibility, and an anchored structural-divergence ratio against a matched null ( $4.7\times$  schema-free,  $2.3\times$  FIBO — structural cross-view methods have signal on one condition and not the other). Exploratorily, per-category verifiable share tracks the graph's grounded-correct lift (Spearman  $\rho = +0.76$ ,  $n = 8$  categories); a prospective test of that link was registered before the reasoning-stratum experiment ran (Appendix F). Every hypothesis in this paper carried its disconfirming outcome before its run; the ledger, registrations, and all artifacts ship in the supplementary material.

**Table 1:** Primary metric (share of the gold answer’s numbers the answer states) by question stratum. Bold marks graph conditions that beat passages with a 95% paired bootstrap interval excluding zero; the reasoning stratum is the only one where that happens, and there it happens on every model.

| Stratum    | Model    | closed book | passages | graph (no schema) | graph (FIBO) | graph (FIBO+pointers) |
|------------|----------|-------------|----------|-------------------|--------------|-----------------------|
| balanced   | gpt-oss  | .306        | .398     | .355              | .384         | .364                  |
|            | MiniMax  | .185        | .484     | .490              | .447         | .502                  |
|            | DeepSeek | .058        | .295     | .221              | .233         | .283                  |
| arithmetic | gpt-oss  | .119        | .530     | .346              | .374         | .367                  |
|            | MiniMax  | .143        | .553     | .451              | .474         | .470                  |
|            | DeepSeek | .054        | .309     | .182              | .226         | .271                  |
| reasoning  | gpt-oss  | .153        | .665     | .709              | <b>.725</b>  | .707                  |
|            | MiniMax  | .072        | .763     | <b>.819</b>       | .817         | .852                  |
|            | DeepSeek | .026        | .467     | <b>.548</b>       | <b>.574</b>  | .634                  |

**Table 2:** Grounded-correct counts (evidence contains the answer and the model answered it), balanced sample, 199 numeric-gold cases.

|               | passages | graph (schema-free) | graph (FIBO) | FIBO+pointers |
|---------------|----------|---------------------|--------------|---------------|
| gpt-oss-120b  | 107      | 111                 | 105          | 112           |
| MiniMax-M2.7  | 124      | 138                 | 127          | <b>146</b>    |
| DeepSeek-V3.1 | 98       | 83                  | 82           | 104           |

## 5 Limitations

Single corpus and domain; three extractor models; passages are a retrieval ceiling, not a retriever; the graph conditions test the graph as serialized context, not as a queryable store (deferred, with the token-budget/cache tension, to future work); “evidence contains the answer” is a numeric-token approximation; per-type strata below  $n = 20$  are reported but underpowered. The registered secondary judge panel is calibrated and corroborating but not load-bearing: all three judges pass the four must-fail controls (identity, swap, scale corruption, refusal) only after two prompt repairs the controls themselves forced; each answer is then graded by the two judges that did not write it. Two-judge agreement reaches 0.79 (pairwise 0.76–0.80), ties stay unresolved, and 86 of 1,800 gradings failed at the API and are counted rather than imputed. The panel’s pattern matches the primary metric’s — graph-condition answers are rarely judged flat-wrong (4–10 per condition against 37 closed-book) and abstain more often — but no claim in this abstract rests on a judge verdict; the primary metric and the evidence-conditional decomposition are deterministic. Full verdict tables ship with the supplementary material.

## 6 Conclusion

Ontology guidance, measured end to end, provides detectability rather than extraction quality; the alignment primitive that decides what “the same fact” means should be provenance, not names; and evaluation that cannot see evidence cannot see what structure is for. The graph’s contribution to answering is not more correct answers — it is answers that are grounded, attributable, and honest about what the evidence cannot support. The value of LLM extraction, measured end to end, is not the quality of its triples but their auditability — and auditability comes from provenance, not from the ontology.

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## GenAI Usage Statement

In this work we used a generative AI coding and research assistant (an LLM-based agent) for core research content: implementing the experiment harnesses and analysis code, refining author-proposed hypotheses into registered form, drafting substantial parts of this manuscript, and executing the pre-registered analyses. All direction — the research questions, the study design choices, the decision to pre-register, and every scope decision — came from the authors, and we have reviewed all AI-assisted work. Every load-bearing number in this paper is machine-verified against frozen run artifacts by a checker included in the supplementary material; the assistant’s code was exercised end-to-end by the reported runs; and all registered verdicts, including the failed ones, are reported as measured. Separately from writing assistance, LLMs are also this study’s *subjects*: the extractor and answering models named in the paper are experimental conditions, not tools of manuscript preparation. We take responsibility for the final content of this work, including text, claims, and artifacts produced with the aid of generative AI.

## A Related work (extended)

**LLM knowledge-graph construction and schema guidance.** Pipelines from GraphRAG [3] to LightRAG [4] and HippoRAG [5] extract entities and relations with an LLM and retrieve over the result; most accept a schema or entity-type list as guidance, and domain deployments reach for standards such as FIBO [1]. What that guidance buys has mostly been assumed rather than isolated: comparisons vary schema together with prompts, models, or chunking. We hold everything but the schema fixed, at two sample scales, and measure agreement, coverage, and violation detectability separately — which is what lets the benefit relocate from generation to validation.

**Evaluating graph-augmented retrieval.** Head-to-head evaluations report win-rates by LLM judge [3] or accuracy on multi-hop suites [6], and disagree with each other by corpus and setup. Two blind spots recur: node identity across systems is resolved by name, and correctness is scored

against gold answers that public-corpus models partly memorize. Both blind spots are our objects of study rather than our instruments — alignment is keyed on source provenance, and every verdict is conditioned on what the served evidence could support.

**Contamination-aware evaluation.** That benchmark answers leak through pretraining is well documented [7], and closed-book baselines are the standard control. Our closed-book gates span 0.06–0.31 across models on the same questions, and the representativeness check shows a gate that passes on a balanced sample can fail on the corpus’s natural mix — so we treat contamination not as a dataset property but as a per-model, per-stratum quantity, and score answers by evidence-conditional decomposition rather than by gold match alone.

**Judging and finance-QA metrics.** Finance QA scores what matters in this domain — the numbers — with execution accuracy under tolerance [8, 9]; general RAG evaluation leans on LLM judges and semantic similarity [10–12], whose position, verbosity, and self-preference biases are documented [13, 14]. Our primary metric is the finance-QA family adapted to prose gold answers; the judge panel is reference-anchored, cross-judging (no model grades its own answers), and calibrated on must-fail controls before its verdicts count.

## B Study design: full detail

### B.1 Dataset and sampling

FinDER pairs terse analyst questions with gold answers and the filing passages that support them (5,703 cases, eight categories). Its own annotations partition the corpus into three strata: *simple* questions (neither annotation), *arithmetic* questions (an operation label such as Division), and *reasoning* questions (flagged as requiring composition, no arithmetic). We run three samples that together cover the grid: a category-balanced 280-case sample (35 per category, seed 42, multi-reference preferred), a 140-case arithmetic-stratum sample (type-proportional), and a 204-case census of the reasoning stratum — a census rather than a sample, so there is no draw to dispute. The balanced sample contains zero arithmetic cases — the multi-reference preference excluded them, since 881 of 883 arithmetic questions carry one reference — which is itself a finding about sample design and the reason the supplement exists.

### B.2 The schema spectrum

Five extraction conditions differ only in the schema handed to the extractor: **A** a schema-free floor (one class, one relation); **B** a hand-written 20-class schema; **C** 70 classes derived from FIBO, corpus-scoped; **D** = C plus FIBO synonyms and abbreviations; **E** = C plus the subsumption hierarchy. Everything else — documents, chunking, extractor models (DeepSeek-V3.1, gpt-oss-120b, MiniMax-M2.7), prompts, property floor, graph store — is held fixed, so schema is the only moving part. Conditions A and C are re-run at 280 cases; each case’s graph is extracted independently per model, giving three *views* per case and condition.

Each condition exists to isolate one belief (Table 3). Condition C’s classes are not hand-picked: every FIBO class is ranked by how many corpus documents mention its label (word-boundary matching, alias-aware so that abbreviation-only classes are not silently discarded), and the top 70 across five FIBO domains are kept — the ontology as the corpus actually uses it, not as its authors organized it. The conditions nest ( $C \subset D$ ,  $C \subset E$ ) so each contrast is attributable to one addition, and the manipulation is verified from the recorded prompts themselves: base system-prompt sizes measure 4.6K (A), 15.4K (B), 19.8K (C), 20.5K (D), 20.5K (E) characters — D’s synonyms add only 724 characters to C, which sharpens the dose–response reading: a very small amount of non-canonical vocabulary produces a large fragmentation effect.

### B.3 Pre-registration

Each run is governed by a registration committed before it starts, stating direction and the outcome that would disconfirm it; exploratory findings are labelled and their registered replications named. The hypothesis ledger and all registrations ship in the supplementary material. We report every registered hypothesis, including the two that failed.

**Table 3:** The schema spectrum: what each condition supplies and what belief it isolates.

|   | The extractor receives            | The belief it tests                 |
|---|-----------------------------------|-------------------------------------|
| A | one class, one relation           | floor: what agreement costs nothing |
| B | 20 hand-written financial classes | a small bespoke schema suffices     |
| C | 70 corpus-scoped FIBO classes     | the standard ontology, as used      |
| D | C + FIBO synonyms/abbreviations   | richer vocabulary converges naming  |
| E | C + the subsumption hierarchy     | structural context improves typing  |

## C The schema spectrum: full results

### C.1 Name agreement falls as the schema grows

Cross-model agreement on entity names is highest under the schema-free floor and falls under every ontology condition (paired A–C difference in per-case comparable-name rate: +0.069, 95% CI [+0.046, +0.091], 280 cases). The mechanism is structural: one declared class gives two extractors nothing to disagree about; seventy classes are seventy naming choices.

### C.2 Content coverage does not rise

The registered hypothesis that FIBO buys content at the price of names — supported at 16 cases — *reversed* at 280: the FIBO condition covers fewer gold-answer numbers than A (−0.036 [−0.067, −0.008]). The ontology has no measured extraction benefit on this corpus.

### C.3 What survives: detectability

Schema violations (missing required properties, undeclared labels, datatype breaks) number 529–1,038 per ontology condition and are mechanically checkable against the declared schema; the schema-free floor offers almost nothing to check. The ontology’s contribution relocates from generation quality to *verification surface* — consistent with keeping heavy ontology reasoning out of the extraction path and using it as an offline validator.

### C.4 Why the synonym condition could not matter

Only 30 of 5,703 questions (0.5%) require vocabulary mapping at all, and the abbreviations questions actually use (rev, mgmt) are analyst shorthand, not FIBO designations. Condition D was measuring a need the corpus does not have — a caution for scoping ontology vocabulary work by measured query demand rather than by ontology completeness.

### The five-point dose–response at scale

The full schema spectrum at 280 cases (name comparable rate, case bootstrap, same cases and models throughout):

**Table 4:** The five-condition dose–response at 280 cases: cross-model name-agreement rate (comparable-key rate) with case-bootstrap 95% intervals. Agreement is not monotone in class count — the hand-written 20-class schema sits below full FIBO — and both synonym and hierarchy additions separate below plain FIBO.

|        | no schema   | FIBO (70)   | FIBO+synonyms | FIBO+hierarchy | hand-written (20) |
|--------|-------------|-------------|---------------|----------------|-------------------|
| rate   | .293        | .211        | .140          | .126           | .118              |
| 95% CI | [.269,.319] | [.191,.232] | [.120,.163]   | [.106,.146]    | [.097,.142]       |

All three registered directions for this curve failed: agreement is not monotone in class count (B’s twenty bespoke classes fragment naming below FIBO’s seventy), and both synonym lists and hierarchy text separate *below* plain FIBO at scale after being indistinguishable at 16 cases. What predicts fragmentation is not how many classes are declared but how much non-canonical lexical material the prompt carries.

## D Provenance-anchored alignment: full results

Two views of the same case hold “the same fact” if we can align their facts. The standard key is the entity name. We instead locate each extracted value at the unique numeric token it came from in the source passage (searching all unit scales, so a mis-scaled extraction still anchors), and align facts that anchor to the same token.

At 280 cases on the schema-free condition, name keys yield 992 comparable fact pairs with 324 disagreements; anchor keys yield 1,656 pairs with 606 disagreements, of which **485 are invisible to any name key** — the views named the fact differently *and* disagreed on its value. On the FIBO condition the multiplier is larger (717 vs. 1,983 pairs; 246 vs. 874 disagreements; 734 name-blind). Both replications were registered in advance. The disagreements the anchor exposes are dominated by unit-scale misreadings: 26.7% of anchored values match their source only after rescaling by  $10^3$  or  $10^6$ .

Two consequences follow. For pipeline builders: entity resolution keyed on names silently discards most of the verification signal that running a second extractor was supposed to buy. For evaluators: cross-model verification has a hard ceiling regardless of key — 87% of facts are held by exactly one view, a share that *rises* with sample size (77.1% at 16 cases) — and agreement-gating trades four correct answers for every wrong answer it blocks (precision  $0.951 \rightarrow 1.000$ , recall  $0.951 \rightarrow 0.764$ ; the registered hypothesis that gating helps was disconfirmed on its own criterion).

## E Answering under five evidence conditions: full results

### E.1 Design

Each question is answered under five evidence conditions — closed book, gold passages, serialized schema-free graph, serialized FIBO graph, and the FIBO graph with provenance pointers — by the same three models at temperature 0. The serializer is one code path; document-hierarchy nodes are excluded because they carry the passages verbatim. Passages are the *ceiling* of retrieval (the gold evidence in full), and the graph conditions serve the union of all three extractor views, the ceiling of what extraction captured. Primary metric: scale-normalized overlap between the gold answer’s numbers and the produced answer’s (the finance-QA standard family [8, 9]); a three-judge cross-judging panel (no judge grades its own model’s answers) covers what number matching cannot.

### E.2 The gate: contamination is real and stratified

Closed-book performance spans 0.058 (DeepSeek) to 0.306 (gpt-oss) on the balanced sample — public filings are memorized to very different degrees — yet passages beat closed book for every model (paired differences  $+0.092$  to  $+0.299$ , all separated). On the arithmetic stratum the gate widens sharply ( $+0.41$  on both models run): *computed* values cannot be recalled, making arithmetic questions the contamination-resistant stratum for evaluating retrieval on public corpora.

### E.3 Naive scoring: text wins or ties; format effects are model properties

Under naive overlap no graph condition beats passages anywhere; DeepSeek loses with graphs on both ( $+0.075$ ,  $+0.062$  toward passages), gpt-oss on the schema-free graph, MiniMax ties. The passages–graph gap itself differs across models beyond resampling (DeepSeek vs. MiniMax:  $+0.081$  [ $+0.019$ ,  $+0.144$ ]) — averaging models erases a real interaction, which is one reason published RAG-vs-GraphRAG aggregates disagree.

### E.4 Evidence-conditional evaluation reverses the reading

Scoring against the gold answer alone cannot distinguish a model that used the evidence from one that remembered the filing. Because we can check mechanically whether the served evidence contains the gold numbers, every verdict decomposes into: grounded-correct, utilization failure, over-refusal, honest abstention, contaminated-correct, and hallucination.

Three facts move (Table 5). The graphs’ grounded yield matches or beats passages on two of three models. Passages carry  $2\text{--}4\times$  the contamination (20–22 vs. 5–10 answers scored correct on numbers absent from the evidence). And DeepSeek’s graph deficit is over-refusal — 55 declines where the

**Table 5:** The full evidence-conditional decomposition on the balanced sample (199 numeric-gold cases): every answer classified by whether the served evidence contained the gold numbers and what the model did. “Grounded” and “over-refusal” condition on evidence that contains the answer; “honest abstention,” “contaminated,” and “hallucination” on evidence that does not.

| Model    | Condition         | grounded | util. fail | over-refusal | honest abst. | contam. | halluc. |
|----------|-------------------|----------|------------|--------------|--------------|---------|---------|
| gpt-oss  | passages          | 107      | 17         | 31           | 9            | 22      | 12      |
| gpt-oss  | graph (no schema) | 111      | 9          | 35           | 25           | 7       | 11      |
| gpt-oss  | graph (FIBO)      | 105      | 9          | 32           | 19           | 15      | 18      |
| gpt-oss  | graph (FIBO+ptr)  | 112      | 10         | 36           | 15           | 10      | 15      |
| MiniMax  | passages          | 124      | 10         | 21           | 15           | 22      | 6       |
| MiniMax  | graph (no schema) | 138      | 6          | 11           | 26           | 10      | 7       |
| MiniMax  | graph (FIBO)      | 127      | 5          | 14           | 27           | 18      | 7       |
| MiniMax  | graph (FIBO+ptr)  | 146      | 3          | 9            | 17           | 11      | 12      |
| DeepSeek | passages          | 98       | 26         | 31           | 11           | 20      | 12      |
| DeepSeek | graph (no schema) | 83       | 17         | 55           | 22           | 5       | 16      |
| DeepSeek | graph (FIBO)      | 82       | 11         | 53           | 24           | 13      | 15      |
| DeepSeek | graph (FIBO+ptr)  | 104      | 22         | 32           | 14           | 10      | 16      |

graph *contains* the answer, vs. 31 with passages — a legibility failure of the serialization for that model, not a failure of the graph’s content. The registered replication on the arithmetic stratum did NOT reproduce the reversal: passages’ grounded-correct count beats every graph condition on all three models there (109/121/96 vs. 80–106), the over-refusal mechanism does reproduce (50/41 graph declines vs. 24 with passages, repaired to 26 by pointers), and contamination is too rare on computed numbers to compare (3–11 per cell, pre-declared uninformative). The decomposition’s reading is therefore stratum-dependent, which the main text carries as a finding (Section 3).

## E.5 Provenance pointers raise accuracy without carrying content

Attaching each anchored value’s source pointer (passage index and character offset — never the source text) was registered to change attribution, not accuracy. The registration was disconfirmed in the interesting direction: accuracy *rose* on MiniMax (+0.055 [+0.012, +0.097]) and DeepSeek (+0.050 [+0.022, +0.080]). Since no answer content was attached, the pointer acts as a trust signal — it marks exactly the numbers that anchored to source — and the pointer condition shows the highest evidence grounding rate (73%/70% of answered numbers present in the served evidence). Mechanical citation checking lands at 42–58% verified, so pointer-following is far from solved; but provenance decoration is a zero-content prompt intervention that buys accuracy.

## E.6 The arithmetic stratum: the original hypothesis meets its test

The graph-favorable hypothesis this study inherited — that structured extraction wins precisely on multi-step, compositional questions — was registered verbatim for the arithmetic supplement, with the honest prior against it. It failed decisively: gold passages beat every graph condition on every model, all intervals separated (+0.079 to +0.184). More telling, the evidence-conditional reversal of Section 6.4 does *not* replicate here: passages’ grounded-correct count beats every graph on all three models (109/121/96 vs. 80–106). On questions that require computing a value from several printed ones, the serialized graph genuinely under-serves even after memorization is stripped — plausibly because the computation needs co-located operands that extraction fragments across nodes. Contamination itself nearly vanishes on this stratum (3–11 answers per cell against 20–22 on the balanced sample), confirming arithmetic questions as the contamination-resistant slice: closed book collapses to 0.05–0.14 while the gate opens four times wider (+0.41).

Two effects did carry over. The over-refusal mechanism reproduces (DeepSeek declines 50/41 graph-condition cases it could have answered, vs. 24 with passages), and the provenance-pointer repair concentrates exactly there: pointers cut its over-refusals to 26 and raise its grounded count from 64 to 82 (+0.045 [+0.013, +0.078]). The pointer effect that the balanced sample located on MiniMax did not reproduce on MiniMax here — the anchors hypothesis, as registered, is disconfirmed in its model-specific form, and what survives is weaker and more honest: pointer-only provenance helps, but which model it helps depends on the stratum.

Together the two strata answer the deployment question differently, which is the finding: *whether a graph is worth serving is a property of the question stratum and the model, not of the representation alone* — and it is measurable in advance, which is what Section 7's profile is for.

## F The verifiability profile and design rules

Owning the full chain — source token, extracted node, served evidence, answer, verdict — lets each failure class be traced to its cause and converted to a rule. Unit-scale misreads (26.7%) argue for anchor-verifying values at write time. Name fragmentation growing with schema size argues for provenance keys in entity resolution and for deploying ontologies as validators rather than extraction guides. The 87% single-view ceiling caps what multi-extractor redundancy can verify — the budget belongs in provenance. Model-specific over-refusal on serialized graphs makes serialization a per-model design surface. And contamination that survives inside evidence conditions makes evidence-conditional evaluation, not closed-book gates alone, the honest default for public-corpus benchmarks.

### F.1 A verifiability profile as a pre-deployment diagnostic

The design decisions this paper evaluates — multi-extractor verification, ontology guardrails, structural alignment — each pay off only under measurable preconditions, and every precondition is computable from one cheap pilot extraction, before any at-scale spend. We consolidate them: the *verifiable share* (facts held by  $\geq 2$  views: 13% here, so cross-view verification has a low ceiling); the *name-blind disagreement rate* (80–84%: how much signal a name-keyed pipeline discards); the *scale-fragility rate* (26.7%: the case for write-time anchor verification); the *guardrail surface* (violations per fact: the ontology's measurable value); per-model *legibility* (over-refusal on serialized evidence); and a structural component measured here for the first time. Anchors give cross-view node identity for free, so structural similarity of *the same* entity across views becomes well-defined: neighborhood agreement of anchor-identified entities is  $4.7\times$  a matched null under the schema-free condition but only  $2.3\times$  under FIBO — even when two views provably hold the same source fact, they embed it in almost disjoint structure (79% and 95% of same-fact pairs share no anchored neighbor at all), and the ontology fragments that structure further. A structural alignment method (SimRank-style or learned) has usable signal only where this ratio clears its null, which makes the ratio a go/no-go gate rather than a score. Validating the full profile as a predictor of downstream grounded-correct lift is deferred to follow-up work.

### Model heterogeneity as a routing argument

Every component above varies across models given identical documents, schema, and prompts, and the variation is large enough to route on. The three extractors' facts overlap on only 13% of anchored values, so per-case extractor selection has measured headroom: an oracle choosing one model per case covers 0.458 of gold-answer numbers against 0.389 for the best fixed model (the oracle's picks split 103/49/47 across the three). Downstream, which evidence *format* a model can use is equally model-specific — the passages-minus-graph gap differs across models beyond resampling, one model over-refuses serialized graphs (55 vs. 24 declines) while another almost never does (2–5), and pointer decoration helps two models and not the third. A multi-agent system serving one representation to all models is therefore leaving measured accuracy on the table. The cautionary half of the argument is measured elsewhere: a learned router over partitioned stores can score below random selection when a miss serves no evidence at all, so the headroom quantified here should be harvested behind a fallback, not instead of one.

## G Registrations and verdicts

Every hypothesis this study registered, in plain language, with its verdict. The identifiers in the left column exist only to cross-reference the dated registration files in the supplementary material, each committed before the run it governs; nothing in the body text depends on them. Exploratory findings are absent from this table by construction — they are labelled in the text and their registered replications appear here.

| Family                                             | Hypothesis (compressed)                                                                                                         | Verdict                                 |
|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| <i>Second sweep (16 cases, five conditions)</i>    |                                                                                                                                 |                                         |
| SW-H1                                              | equal property floor removes period effect, not agreement                                                                       | held                                    |
| SW-H2                                              | hierarchy (E) raises agreement over plain FIBO                                                                                  | not separated                           |
| SW-H3                                              | FIBO content advantage survives equal slots                                                                                     | held at 16; see S2-H4                   |
| SW-H4                                              | no ontology condition escapes name fragmentation                                                                                | held                                    |
| <i>Scale-up, schema-free (280 cases)</i>           |                                                                                                                                 |                                         |
| SC-H1                                              | verifiable-fact shortage is sample-size-bound                                                                                   | held (6→123)                            |
| SC-H2                                              | agreement-gated serving beats serve-always                                                                                      | <b>disconfirmed</b>                     |
| SC-H3                                              | single-view ceiling does not move                                                                                               | held (rose to 87%)                      |
| SC-H4                                              | anchor advantage survives scale (schema-free condition)                                                                         | held                                    |
| <i>FIBO at scale (280 cases)</i>                   |                                                                                                                                 |                                         |
| S2-H1                                              | anchor advantage replicates on the FIBO condition                                                                               | held                                    |
| S2-H2                                              | name-agreement gap persists, paired                                                                                             | held                                    |
| S2-H3                                              | scale-error rate stable ( $\approx$ a quarter)                                                                                  | held (26.7%)                            |
| S2-H4                                              | FIBO content advantage survives scale                                                                                           | <b>disconfirmed, reversed</b>           |
| <i>Answering, balanced sample</i>                  |                                                                                                                                 |                                         |
| AN-H1                                              | models cannot already answer (gate)                                                                                             | held 3/3                                |
| AN-H2                                              | graph does not beat its source text                                                                                             | held                                    |
| AN-H3                                              | passages—graph gap is a model property                                                                                          | held (1 of 3 pairs separated)           |
| AN-H4                                              | anchors change attribution, not accuracy                                                                                        | <b>disconfirmed</b> on 2/3              |
| <i>Arithmetic supplement</i>                       |                                                                                                                                 |                                         |
| AR-H1                                              | gate holds on arithmetic                                                                                                        | held (+0.41)                            |
| AR-H2                                              | a graph condition beats passages (the original claim)                                                                           | <b>disconfirmed</b>                     |
| AR-H4                                              | anchor effect reproduces on MiniMax                                                                                             | <b>disconfirmed</b> ; moved to DeepSeek |
| AR-EC1                                             | grounded-correct reversal replicates                                                                                            | <b>disconfirmed</b> (stratum-dependent) |
| AR-EC2                                             | contamination asymmetry                                                                                                         | uninformative (pre-declared)            |
| AR-EC3                                             | over-refusal concentrates on graphs (DeepSeek)                                                                                  | held                                    |
| <i>Representativeness (natural mix, graphless)</i> |                                                                                                                                 |                                         |
| RP-H1                                              | gate replicates per model                                                                                                       | partial: MiniMax yes, gpt-oss <b>no</b> |
| RP-H2                                              | balanced design did not distort gate size                                                                                       | held (intervals overlap)                |
| <i>Multi-agent adjudication</i>                    |                                                                                                                                 |                                         |
| MA-H1/H3                                           | provenance beats deliberation / beats pick-only rule<br>— all 1,482 disagreements are unit-scale splits; audited before scoring | <b>void</b> : ground truth defective    |

## H Scoring-metric sensitivity

The conclusions do not depend on the primary metric’s choice, and the one place they would have is itself informative. Rescoring every balanced-sample answer with token-level F1 (the TAT-QA-style metric) preserves the gate on both models tested (passages – closed book +0.071 and +0.082, both separated) but flips two comparisons in directions the metric’s own biases predict: passages separates above graphs everywhere (gold answers are written from the passages, so surface overlap is a gift to the passages condition), and the pointer condition scores *below* plain graph on gpt-oss (−0.016, separated) — because citation markers are tokens the gold answer does not contain, i.e., token F1 penalizes the act of citing. A metric that rewards passage-copying and punishes attribution cannot be primary in a study about verifiability; it is reported here so the reader can see exactly what it would have changed.

Two tokenizer defects were caught by a four-call smoke test before any scored run, and fixed symmetrically for gold and produced text: a hyphen after an alphanumeric read as a minus sign (“SP 800-171” yielding −171), and parenthesized enumeration markers “(1)...(6)” counted as numbers. The scoring tokenizer is separate from the anchoring tokenizer, which is left untouched because the anchors were located with it.